UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

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Interconnection of Large Loads to the	)	Docket No. RM26-4-000
Interstate Transmission System	)	

COMMENTS OF CLEARPATH, INC.

ClearPath, Inc. ("ClearPath") respectfully submits these comments in response to the Federal Energy Regulatory Commission's ("FERC" or "Commission") Advanced Notice of Proposed Rulemaking ("ANOPR") issued on October 27th, 2025, for Ensuring the Timely and Orderly Interconnection of Large Loads. ClearPath's mission is to accelerate American innovation to reduce global energy emissions. To advance that mission, we develop cutting-edge policy solutions on clean energy and clean manufacturing innovation. An entrepreneurial, strategic nonprofit, ClearPath (501(c)(3)) collaborates with public and private sector stakeholders on innovations in carbon capture, nuclear energy, natural gas, geothermal, hydropower, energy storage and clean manufacturing to enable private-sector deployment of critical technologies.

I. Communications

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II. Background

ClearPath supports the Department of Energy's (DOE) focus on advancing the United States' global leadership in Artificial Intelligence (AI) and revitalizing the domestic manufacturing base. Regulatory reform can be critical for providing the private sector with certainty and clarity for capital investments while enabling innovative technologies and business approaches to thrive unencumbered.

By 2030, an estimated $7 trillion will be invested into data centers globally, with 40% of this investment in the United States.[1] Investments in new or expanded domestic manufacturing facilities have skyrocketed, with $540 billion invested in the domestic semiconductor supply chains in response to the Creating Helpful Incentives to Produce Semiconductors (CHIPS) and Science Act.[2] In addition, over $290 billion has been invested in clean energy manufacturing,[3] and over $7 trillion of committed investment in U.S. manufacturing and innovation has been made as a result of the Trump Administration policies.[4] Meanwhile, electric sector spending could reach $1.4 trillion by 2030 to support this demand growth, in addition to modernizing its distribution and transmission systems.[5]

To realize this generational opportunity for economic growth, technological leadership, and grid development, regulatory predictability and certainty must be established, and innovative technologies and business models must be enabled, not hindered. For novel technologies that enable large loads to operate as dispatchable, removing market barriers is essential to allowing these business models to operate economically and reduce their impact on the broader system through flexible operations.

Historically, all loads have been primarily viewed as static consumers of energy, with little to no attributes that contribute to reliable grid operations. This is no longer the case. Today, we have large load options and technologies that can connect to the transmission grid, responsively manage load, provide storage opportunities, and provide balancing services to the grid.

Dispatchable loads are controllable energy demand that quickly start and stop consuming power, and for some technologies, can store energy. The most flexible of these technologies can follow energy price signals, adjusting their loads up and down to consume power when prices are lower and avoiding periods with high prices or emergency conditions. Dispatchable large loads can

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1 https://www.mckinsey.com/industries/public-sector/our-insights/the-data-center-balance-how-us-states-can-navigate-the-opportunities-and-challenges
2 https://www.csis.org/analysis/innovation-lightbulb-tracking-chips-act-incentives
3 https://cleaneconomytracker.org/
4 https://www.whitehouse.gov/articles/2025/08/trump-effect-a-running-list-of-new-u-s-investment-in-president-trumps-second-term
5 https://www.deloitte.com/us/en/insights/industry/power-and-utilities/funding-growth-in-us-power-sector.html

therefore avoid peak demand periods, reducing the need for additional infrastructure build-out, such as transmission facilities or additional capacity resources. Large loads that are also capable of storing power will have much lower capacity factors, allowing for a greater degree of operational flexibility that can reduce system impacts and enable faster interconnection compared to inflexible large loads.

The first example of a dispatchable large load is thermal energy storage (TES), also known as thermal batteries. This is a class of storage technologies that converts electricity into on-demand process heat for industrial customers.[6] TES can also reconvert heat to power, for use by industrial customers or resale to utilities or in wholesale markets as a long-duration storage and capacity resource. TES can be directly connected to the transmission system, located behind the customer's meter, and withdraw power from the grid or a directly connected generator. This heat can be utilized immediately by the industrial load or stored for 1-2 days, or even longer, depending on the application. Commercially available thermal batteries, such as Rondo,[7] Brenmiller Energy,[8] and Antora,[9] can supply heat at temperatures reaching 1,380°F, making them compatible with 75% of industrial heating applications.[10] These technologies can decrease costs for industrial and manufacturing facilities due to their flexible operations, which allow TES to withdraw electricity when it produces heat more cost-effectively than alternative heat sources. TES can also provide grid balancing services because they can be scheduled in advance or react rapidly, on the order of milliseconds to minutes.[11] As a result, individual utilities, balancing authorities, regional transmission operators (RTOs), and independent system operators (ISOs) can draw on TES to deliver reliability and ancillary services to manage outages and grid ramping needs. Providing regulatory predictability for interconnection and aligning operating capabilities with market access can support American industrial cost competitiveness in sectors such as ethanol and chemical manufacturing.

The second example of a dispatchable large load is electrolyzers. This includes the category of technology that splits water into oxygen and hydrogen using an anode and a cathode separated by an electrolyte.[12] These technologies can pull power from the grid during low-price periods to create hydrogen that can be used for petroleum refining, fertilizer production, or as a fuel. However, cost-competitive electrolytic hydrogen production requires the ability to purchase electricity at wholesale prices in order to advance American industrial competitiveness.

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6 https://www.brattle.com/wp-content/uploads/2023/10/Thermal-Batteries-Opportunities-to-Accelerate-Decarbonization-of-Industrial-Heat.pdffacilities'canby allowing them to withdraw electricity only when it becomesonly to withdraw electricityf

7 https://www.rondo.comthe total outputrequiring/

8 https://bren-energy.com/

9 https://www.antora.comcompetitivenessto/

10 https://www.brattle.com/wp-content/uploads/2023/10/Thermal-B.atteries-Opportunities-to-Accelerate-Decarbonization-of-Industrial-Heat.pdf

11 *Ibid*

12 https://www.energy.gov/eere/fuelcells/hydrogen-production-elecmost significantkWelectric-intensiveincentivizetrolysiones

A third example of dispatchable large loads is the Electric Arc Furnace (EAF). EAFs are the dominant method of steel production in the U.S., accounting for 71.8% of total production in 2024.[13] They are energy-intensive customers, and as an example, can be rated at 120 MW of load.[14] At the same time, EAFs are capable of curtailing and shifting loads quickly[15] and have an incentive to reduce electricity consumption to enhance competitiveness, as energy accounts for 20-40% of steel production costs.[16]

Aligning market access, energy pricing signals, and interconnection procedures with the technological capabilities and business models of dispatchable large loads is critical for achieving just and reasonable rates, terms, and conditions that are not unduly discriminatory or preferential, and fosters private sector innovation and competitiveness. ClearPath applauds the Administration's leadership in initiating this proceeding to determine a path forward that unleashes American energy resources to power our global leadership in AI and manufacturing while ensuring that state authority is respected.

III. Comments

A. Principle Three: FERC should better integrate transmission development with interconnection

Timely and reliable interconnection of large loads is critical for our economy and national security. ClearPath strongly supports the ANOPR's proposal to jointly study supply-side interconnection customers and large load interconnection customers to the extent practicable. Two of the most important pieces of information for interconnection customers to make informed investment decisions are the cost and timeframe for interconnection. As a result, developers of both generation and large loads often submit multiple interconnection requests as a means of price discovery and time-to-grid discovery. This creates unmanageable queue volumes, inflated demand and supply forecasts, and significant delays to the interconnection process due to projects withdrawing late into the process when they discover their project is not economically feasible due to costs or energization timelines.

Coordinating interconnection studies for large loads with supply-side interconnection can enable the identification of more cost-effective transmission solutions and ensure that all benefiting customers pay, whether they are load or supply customers. This coordination can also improve a project's commercial readiness because large loads that are interested in offtake agreements with generation will be processed on similar timelines, which reduces uncertainty for both parties.

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[13] https://worldsteel.org/wp-content/uploads/World-Steel-in-Ficompetitivenessgures-2025.pdf

[14] https://psc.ky.gov/tariffs/electric/East%20Kentucky%20Power%20Cooperative%2C%20Inc/Contracts/Owen%20Electric/Nucor%20Steel%20Gallatin/2013-06-02_Agreement%20for%20Electric%20Service.pdf?

[15] https://psc.ky.gov/order_vault/Orders_2025/202500208_08192025.pdf

[16] https://worldsteel.org/wp-content/uploads/Fact-sheet-energy-in-the-steel-industry-2021-1.pdf

In addition to coordinating interconnection studies, ClearPath recommends that FERC consider how the coordination and integration of transmission planning and interconnection processes – for both large loads and generation – also facilitates just and reasonable wholesale rates. Proactively planning transmission can increase transparency on the cost and timing of interconnection and identify cost-effective transmission solutions. In particular, when transmission is proactively planned, the costs for interconnection are provided to interconnection customers upfront, and developers have greater clarity on when they can interconnect, reducing uncertainty and the speculative behavior that increases queue timelines. There are nascent efforts on this path.

The California Independent System Operator (CAISO) has implemented reforms to its interconnection process that focus this process on areas where transmission capacity is available and where transmission is being developed.[17] The Southwest Power Pool (SPP) has submitted its Consolidated Planning Process (CPP) for FERC approval, which aligns transmission planning with interconnection to identify more optimal infrastructure solutions that provide multiple benefits and provide a predictable and transparent framework for allocating those costs to the beneficiaries.[18] An Open Season process for developing transmission, such as Bonneville Power Administration's Network Open Season,[19] or the Competitive Renewable Energy Zones developed in Texas[20] are two examples of proactive transmission planning approaches that provide a scalable solution for bringing projects online in a more predictable manner that transparently allocates costs to customers commensurate with the benefits they receive.

B. Principle Five: Large Loads must be studied according to their planned operations and engineering designs

ClearPath recommends that FERC require transmission providers to study large loads based on their known operating characteristics to increase the accuracy of these studies, avoid unnecessary network upgrades that can slow interconnection, and reduce excessive costs. In Order 2023, FERC requires transmission providers to study electric storage resources' according to their planned operating assumptions and only for withdrawals of energy.[21] We recommend that FERC adopt a similar approach for studying large load withdrawals and/or injections. This advances just and reasonable rates by avoiding overestimation of the impacts of large loads through study assumptions that depict worst-case scenarios, such as curtailable large loads withdrawing at full capacity during peak demand periods, or conflicts with installed control technologies or system

[17] https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20240930-3094&optimized=false

[18] https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20251103-5149&optimized=false&sid=7c8ec448-abb4-495d-b95e-f5d1266d32e4

[19] https://www.nrg.com/assets/documents/energy-policy/grid-strategies-electric-network-open-seasons080924.pdf

[20] https://www.nrc.gov/docs/ML0914/ML091420467.pdf

[21] https://www.ferc.gov/media/order-no-2023

protection facilities. This also reduces undue discriminatory or preferential barriers to large loads that employ technological innovations, enabling them to be curtailable or dispatchable.

C. Principle Seven: FERC should remove market barriers to economically-efficient, flexible operations

FERC's pro forma interconnection procedures for large loads should offer different interconnection service options to facilitate customer choice, particularly for customers with operating capabilities such as dispatchability or curtailability. These could be similar to the interconnection service options available to generation and storage interconnection customers, including network resource integration service (NRIS), energy resource integration service (ERIS), provisional ERIS, and surplus interconnection service. Customers seeking interconnection at different levels or through various arrangements, such as co-location, should be studied separately to the maximum extent practicable, enabling faster interconnection when possible. For example, CAISO's interconnection process for generation and storage prioritizes projects that can reliably and safely interconnect into the system sooner over projects that have long lead times for their network upgrades.[22]

FERC's interconnection procedures for large loads must enable innovations that can increase the dispatchability of large loads by providing access to markets that unlock the full potential for flexible operations that can expedite interconnection. Federal Power Act (FPA) section 201 makes clear that Federal regulation extends to "the sale of such energy at wholesale in interstate commerce" and defines this in section 201(d) as "a sale of *electric* energy to any person for resale"[23] (*emphasis* added). Further, FERC has exercised its rate jurisdiction under FPA section 205 for matters materially affecting rates, terms and conditions of jurisdictional service. For dispatchable large loads, there is the potential for hybrid engagement by large loads in wholesale markets. For example, storage and other technologies that may be associated with a large load may sell energy back into the grid or provide ancillary services supporting grid operations. In the case of TES, these large load technologies are converting this electricity into other products, such as heat, that are then consumed or sold to a large load. Other dispatchable large loads discussed here, including hydrogen electrolyzers and electric arc furnaces, consume energy for developing other products. The rates or charges for such wholesale market participation by large load technologies must be considered subject to FERC jurisdiction under FPA section 205(a).[24]

Any interconnection agreement for large load cannot prohibit the load from engaging in jurisdictional wholesale market activities such as the provision of energy (e.g., discharge of energy from storage into the grid) or ancillary services, including voltage support and other

[22] https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20240801-5172&optimized=false
[23] https://www.ferc.gov/sites/default/files/2021-04/federal_power_act.pdf
[24] https://www.ferc.gov/sites/default/files/2021-04/federal_power_act.pdf

similar support for balancing the grid. To this end, FERC must ensure that dispatchable large loads, in addition to co-located generating facilities, are fully compensated for providing energy and ancillary services to the grid. In response to FERC's order directing reports on the Modernizing Wholesale Electricity Market Design, each of the Regional Transmission Organizations and Independent System Operators submitted comments that underscored the importance of flexible resources for supporting operational reliability on their systems and the new market products they are developing to incentivize and compensate flexible resources for the services they provide.[25] Dispatchable large load facilities can support operational reliability by providing ancillary services, such as reserve, regulation, and ramping products, through controllable curtailment and load-following operations. FERC should foster, not hinder, these innovative technologies by supporting market access and cost-allocation frameworks that allow dispatchable large loads to support operational reliability in an economically efficient manner.

We also recognize that, currently, there is no regulatory framework that provides dispatchable large loads access to purchasing energy at wholesale energy prices. The practical reality is that a large load interconnecting at a transmission voltage level also may be authorized under applicable state siting and permitting laws to construct, own, and operate the step-down facilities and feeders from the point of interconnection to its own facilities. Further, the large load interconnection will be metered and, in organized markets, likely represent a pricing node with its own locational pricing. FERC's finding that "the interconnection of large loads to the transmission system falls under a practice directly affecting Commission-jurisdictional wholesale electricity rates" expands beyond the interconnection process to include these large load impacts on price formation in wholesale energy markets. This will be particularly true of large loads that agree to be curtailable or dispatchable as a condition of their interconnection agreement, as these changes in demand will have direct effects on wholesale prices.

Action on direct access of large loads to wholesale energy markets is a necessary innovation that lines up with prior Commission actions recognizing the changing nature of grid technologies and capabilities. Passed in 2011, Order 745 established rules for compensating demand response resources, which are technologies that sell demand response services by "reducing consumption of electric energy from their expected levels".[26] This opened the door for price-responsive demand to be compensated at the locational marginal prices, provided it passes the net benefit test, in addition to existing demand response programs designed and administered to encourage curtailable demand to respond with decreased consumption during reliability or emergency conditions. Loads that are curtailable can incur economic losses associated with reducing their electricity consumption. At the same time, price-responsive demand must pass the net benefit test to clear the day-ahead or real-time market. Order 745 does not permit demand response

[25] https://www.ferc.gov/media/ad21-10-000
[26] https://www.ferc.gov/sites/default/files/2020-06/Order-745.pdf

resources to purchase power at the wholesale rate; instead, they are only compensated at the wholesale rate for reduced consumption.

In 2018, FERC approved Order 841, which allows electric storage resources to participate in wholesale markets as both sellers and buyers. Electric storage resources are defined as "a resource capable of receiving electric energy from the grid and storing it for later injection of *electric* energy back to the grid" (*emphasis* added).[27] To the extent that some dispatchable load only receives energy from the grid, Order 841 does not facilitate their ability to purchase electricity at wholesale rates. Additionally, TES can optimize its operations to convert electricity into heat for resale or reconvert the heat to electricity. Only the latter transaction would fall under FERC's jurisdiction under Order 841, creating additional regulatory and economic uncertainty.

These were followed in 2020 by Order 2222, which removed barriers to entry for aggregations of distributed energy resources to participate in wholesale markets. This rule defines a distributed energy resource (DER) as any resource behind a customer meter and a DER aggregator as "an entity that aggregates one or more distributed energy resources".[28] FERC required RTOs and ISOs to establish bidding parameters for DERs that require purchasing power to operate, such as electric vehicles and electric battery storage. Because Order 2222 is extending the market access afforded by Order 841 and compensation mechanisms offered by Order 745, dispatchable large loads face the same barriers to accessing wholesale power rates. Further, while FERC declined to set a maximum size for individual resources in an aggregation, Order 2222 required each RTO and ISO to submit maximum size limits or provide justification for not establishing limits in their compliance filings. As a result, PJM adopted a maximum size of 5 MW[29] and CAISO adopted 1 MW[30] for DERs, which further excludes the potential participation of large loads as aggregators with wholesale market access.

In a similar vein, progress has been made at the state level to devise tariffs and market structures to deliver wholesale power access to large controllable loads. In July 2025, Otter Tail Power Company received approval for a 20-year Electric Service Agreement (ESA), with deviations from the standard tariff schedule, from the South Dakota Public Utility Commission.[31] The ESA governs terms of power delivery for the controllable installation of an Antora TES facility, which will convert electricity into dispatchable heat for an industrial facility.[32] While details of the key deviations are confidential, the ESA includes a dual-tariff structure to serve the load, with a specific Thermal Market Energy Pricing Rider for the non-firm load. The Rider more closely reflects wholesale power pricing by providing day-ahead prices based on the hourly Midcontinent Independent System Operator (MISO) locational marginal price at the facility's

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[27] https://www.ferc.gov/media/order-no-841
[28] https://www.ferc.gov/sites/default/files/2020-09/E-1_0.pdf
[29] https://www.pjm.com/-/media/DotCom/committees-groups/subcommittees/disrs/postings/ferc-order-no-2222-overview.pdf
[30] https://www.caiso.com/Documents/Aug15-2022-ComplianceFiling-FERC-Order-No-2222-ER21-2455.pdf
[31] https://puc.sd.gov/commission/dockets/electric/2025/el25-015/EL25-015ServiceAgreement.pdf
[32] *Ibid*

commercial pricing node, Network Integration Transmission Service costs, and any other MISO charges incurred by Otter Tail to serve this load.[33] Approval of the ESA, including the Rider, highlights the South Dakota PUC's recognition that greater access to wholesale prices more accurately aligns the operating capabilities of dispatchable loads with the costs they cause on the system and supports a growing industrial base. Despite this progress, a jurisdiction-by-jurisdiction approach adds additional cost, time, and uncertainty to project development. Addressing the issue of wholesale energy price access for large loads connecting at the transmission level is therefore critical to revitalizing domestic manufacturing using innovative technologies like EAF, electrolyzers, and TES.

Simply, adopting large load interconnection procedures alone is not enough. We urge the Commission to take into consideration how its large load interconnection procedures and other FERC initiatives can ensure accessibility of large load access to wholesale energy markets.

D. Principles Eleven and Twelve: FERC should align costs with system use and impact

ClearPath recommends that FERC establish clear guardrails for how utilities serving large loads allocate costs for transmission and ancillary services to prevent cost-shifting or free riding, rather than prescribing a particular methodology. This approach will ensure that innovative approaches to cost allocation and market design, particularly for ancillary services, can continue to reflect reliability needs and procure those services through efficient market solutions.

FERC should support approaches to cost-allocation for large load interconnections that unlock the flexible capabilities of co-located, hybrid, and dispatchable large load facilities to support operational reliability. Cost-allocation of network upgrades that combine volumetric and coincident peak demand charges can incentivize flexible large loads to avoid system peaks, therefore reducing system impacts and the need for additional infrastructure investments. Finding the right balance between volumetric and coincident peak demand charges could lead to more economically efficient outcomes while ensuring that large loads pay for their use of the system.

Several utilities have piloted or implemented rate structures that leverage and incentivize EAF load controllability and flexibility, recognizing the grid benefits, while EAFs have valued the lower power costs from interruptible tariffs. As an example, utilities in Seattle,[34] North Carolina,[35] and Kentucky[36] have implemented tariffs with EAF customers, including specific

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33 https://puc.sd.gov/commission/tariffs/Electric/ottertail/section14/16.pdf#:~:text=AVAILABILITY:%20This%20rider%20is%20available%20on%20a,that%20is%20not%20oefficientwned%20by%20the%20Company

34 https://clerk.seattle.gov/search/ordinances/121416

35 https://www.dominionenergy.com/-/media/content/rates-and-tariffs/pdfs/north-carolina/large-business/schedule-ns.pdf

36 https://psc.ky.gov/tariffs/electric/East%20Kentucky%20Power%20Cooperative%2C%20Inc/Contracts/Owen%20Electric/Nucor%20Steel%20Gallatin/2013-06-02_Agreement%20for%20Electric%20Service.pdf

curtailment and interruptibility provisions that specify interruptible load hours, compensation, advance notice requirements, and "buy-through" clauses that allow EAFs to avoid curtailment and continue to purchase power. In another example, steel manufacturers in Ohio took advantage of transmission critical peak pricing pilot based on Network Service Peak Load, the metered demand coincident with the single zonal peak load hour for a given year, instead of a flat monthly demand charge based on 15 - 30 minute peak facility demand.[37] This regulatory framework incentivized EAFs to curtail load during critical peak demand and shift production, leading to a reduction of peak loads by 91%.[38]

IV. CONCLUSION

ClearPath respectfully submits these comments to the Commission for its consideration.

Respectfully submitted,

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[37] https://www.aceee.org/sites/default/files/pdfs/ssi21/panel-4/Seryak.pdf?
[38] *Ibid*